

Taraba State 5 km Road Buffer & School Proximity Analysis

Project Overview

This project uses QGIS to examine the proximity of schools to major road networks in Taraba State, Nigeria. A 5 km buffer was created around major roads, and schools located within the buffer were identified using spatial selection and overlay operations.

The project demonstrates practical application of vector data processing, clipping, buffer analysis, spatial selection, and cartographic layout design.

Objective

To identify schools in Taraba State that are located within 5 km of major roads and visualize their spatial distribution.

Interpretation: Schools within the 5 km buffer represent schools in relatively close proximity to major road networks. The buffer itself does not establish that these schools are safe or unsafe it is primarily a measure of road proximity/accessibility.

Data Used

- Nigeria administrative boundaries - vector layer
- Major roads in Nigeria - vector layer
- Schools in Nigeria - vector layer
- Area of interest: Taraba State, Nigeria

Methodology

1. Imported the Nigeria administrative boundaries vector layer into QGIS.
2. Selected and clipped Taraba State from the Nigeria administrative boundary layer to create the study area.
3. Imported the major roads of Nigeria vector dataset.
4. Clipped the road layer to retain only the major roads located within Taraba State.
5. Created a 5 km buffer around the major roads in Taraba State using QGIS geoprocessing tools.
6. Imported the schools in Nigeria vector dataset.
7. Clipped the school layer to retain schools within Taraba State.
8. Used spatial selection based on the 5 km road buffer to identify schools located within the buffered area.
9. Visualized the roads, 5 km buffer, and selected schools on the final map.
10. Prepared a final print layout containing the map title, legend, north arrow, and scale bar.

QGIS Techniques

- Vector data import
- Layer clipping
- Buffer analysis

- Spatial selection
- CRS Management
- Reprojection
- Proximity analysis

Key Output

A map showing major roads in Taraba State, the 5 km road buffer, and schools located within the buffer.

Skills Demonstrated

QGIS | Spatial Analysis | Buffer Analysis | Vector Data Processing | Proximity Analysis | Spatial Selection | Reprojection | Map Layout Design

Project Significance

The workflow can be applied to real-world questions involving accessibility, proximity to transportation networks, service coverage, and location-based planning.